

Ionic Bonding and Formula Units

Independent Application Answer Sheet - Notebook Evidence

Student name: _____ Date: _____

Directions

Open PhET Build an Atom, select the Atom screen, and complete the three ion models shown below. Then use the charge-balance rule to write each formula unit. Submit this completed PDF in U3 L2 Notebook Evidence Submission.

Part A - Build and record ions

Model	Protons	Electrons	Net charge	Evidence
Sodium ion, Na ⁺	11	10		
Chloride ion, Cl ⁻	17	18		
Magnesium ion, Mg ²⁺	12	10		

Part B - Balance charges to write formula units

Ions	Least whole-number ratio	Charge check	Formula unit
Na ⁺ and Cl ⁻			
Mg ²⁺ and Cl ⁻			
Al ³⁺ and O ²⁻			

Explanation

Explain why an ionic formula is a simplest whole-number ratio rather than a separate molecule. Use the terms ions, total charge, formula unit, and lattice.

Correction rule: If either total charge is not zero, change the ratio and recalculate before submitting.